

1.11 Vector and matrix calculus

1.11.1 Gradients of linear and quadratic functions

For a differentiable scalar function f of $\mathbf{x} \in \mathbb{R}^n$, the **gradient** $\partial f / \partial \mathbf{x}$ is the column vector with entries $\partial f / \partial x_i$, and the **Hessian** $\partial^2 f / \partial \mathbf{x} \partial \mathbf{x}^\top$ is the $n \times n$ matrix of second derivatives. For a vector-valued $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$, the $m \times n$ *Jacobian* $\partial \mathbf{g} / \partial \mathbf{x}^\top$ has (i, j) entry $\partial g_i / \partial x_j$.

Theorem 1.11.1 — Derivatives of linear and quadratic forms

Let $\mathbf{a} \in \mathbb{R}^n$, let $\mathbf{A}$ be $n \times n$ and $\mathbf{B}$ be $m \times n$. Then

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{B}\mathbf{x})}{\partial \mathbf{x}^\top} = \mathbf{B}, \quad \frac{\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}, \quad \frac{\partial^2(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{A} + \mathbf{A}^\top.$$

In particular $\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x}) / \partial \mathbf{x} = 2\mathbf{A}\mathbf{x}$ when $\mathbf{A}$ is symmetric.

Proof. The first two are immediate from $\mathbf{a}^\top \mathbf{x} = \sum_i a_i x_i$ and $(\mathbf{B}\mathbf{x})_i = \sum_j b_{ij} x_j$. For the quadratic form, expand

$$(\mathbf{x} + \mathbf{h})^\top \mathbf{A}(\mathbf{x} + \mathbf{h}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{h}^\top (\mathbf{A} + \mathbf{A}^\top)\mathbf{x} + \mathbf{h}^\top \mathbf{A}\mathbf{h}.$$

The middle term is linear in $\mathbf{h}$ and the last is $O(\|\mathbf{h}\|^2)$, so $(\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$ is the gradient. The gradient is linear in $\mathbf{x}$ with matrix $\mathbf{A} + \mathbf{A}^\top$, which is the Hessian by the second formula. ■

Example. (The least squares criterion)

Let $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{b} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\mathbf{b}$. By Theorem 1.11.1,

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{b} = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}), \quad \frac{\partial^2 S}{\partial \mathbf{b} \partial \mathbf{b}^\top} = 2\mathbf{X}^\top \mathbf{X}.$$

Setting the gradient to zero gives the normal equations $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$. The Hessian is nonnegative definite, so S is convex and every solution is a global minimizer. The next result proves this without calculus, and covers the rank-deficient case.

Proposition 1.11.2 — Minimizing a quadratic function

Let $\mathbf{A}$ be $n \times n$ nonnegative definite, $\mathbf{b} \in \mathbb{R}^n$ and $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$. If $\mathbf{b} \in \mathcal{C}(\mathbf{A})$, the minimizers of f are exactly the solutions of $\mathbf{A}\mathbf{x} = \mathbf{b}$, and the minimum value is $-\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$ for any generalized inverse $\mathbf{A}^-$. If $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$, then f is unbounded below.

Proof. Let $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ and let $\mathbf{x}_0$ solve $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$. Completing the square,

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{A}(\mathbf{x} - \mathbf{x}_0) - \mathbf{x}_0^\top \mathbf{A}\mathbf{x}_0.$$

The first term is nonnegative, and by Proposition 1.7.5(b) it is zero iff $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$, that is, iff $\mathbf{Ax} = \mathbf{b}$. The minimum is $-\mathbf{x}_0^\top \mathbf{Ax}_0 = -\mathbf{x}_0^\top \mathbf{AA}^\top \mathbf{Ax}_0 = -\mathbf{b}^\top \mathbf{A}^\dagger \mathbf{b}$. If $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$, write $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$ with $\mathbf{b}_1 \in \mathcal{C}(\mathbf{A})$ and $\mathbf{0} \neq \mathbf{b}_2 \in \mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$ (Proposition 1.5.1). Then $f(t\mathbf{b}_2) = -2t\|\mathbf{b}_2\|^2 \rightarrow -\infty$ as $t \rightarrow \infty$.

■

With $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$, the proposition gives least squares for every model matrix: the minimizers are the solutions of the normal equations, and the minimum of S is $\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^\dagger \mathbf{X}^\top \mathbf{y}$.

1.11.2 Constrained minimization

To minimize a differentiable $f(\mathbf{x})$ subject to linear constraints $\mathbf{Hx} = \mathbf{h}$, with $\mathbf{H}$ of size $q \times n$, the method of **Lagrange multipliers** looks for stationary points of $f(\mathbf{x}) + 2\boldsymbol{\theta}^\top (\mathbf{Hx} - \mathbf{h})$ in $(\mathbf{x}, \boldsymbol{\theta})$. The factor 2 is only for convenience. For quadratic f the stationary point can be written down and checked directly.

Proposition 1.11.3

Let $\mathbf{A}$ be positive definite, $\mathbf{H}$ of size $q \times n$ with full row rank, and $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{Ax} - 2\mathbf{b}^\top \mathbf{x}$. The unique minimizer of f subject to $\mathbf{Hx} = \mathbf{h}$ is

$$\mathbf{x}_* = \mathbf{x}_0 - \mathbf{A}^{-1} \mathbf{H}^\top (\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{Hx}_0 - \mathbf{h}), \quad \mathbf{x}_0 = \mathbf{A}^{-1} \mathbf{b}.$$

Proof. $\mathbf{HA}^{-1}\mathbf{H}^\top$ is positive definite by Proposition 1.7.5(c,e), so $\mathbf{x}_*$ is well defined, and $\mathbf{Hx}_* = \mathbf{h}$ by direct substitution. Stationarity of the Lagrangian gives $2\mathbf{Ax} - 2\mathbf{b} + 2\mathbf{H}^\top \boldsymbol{\theta} = \mathbf{0}$. Indeed $\mathbf{Ax}_* - \mathbf{b} = -\mathbf{H}^\top \boldsymbol{\theta}_*$ with $\boldsymbol{\theta}_* = (\mathbf{HA}^{-1}\mathbf{H}^\top)^{-1} (\mathbf{Hx}_0 - \mathbf{h})$. For any feasible $\mathbf{x}$, put $\mathbf{d} = \mathbf{x} - \mathbf{x}_*$, so that $\mathbf{Hd} = \mathbf{0}$. Then

$$f(\mathbf{x}) = f(\mathbf{x}_*) + 2\mathbf{d}^\top (\mathbf{Ax}_* - \mathbf{b}) + \mathbf{d}^\top \mathbf{Ad} = f(\mathbf{x}_*) - 2(\mathbf{Hd})^\top \boldsymbol{\theta}_* + \mathbf{d}^\top \mathbf{Ad} = f(\mathbf{x}_*) + \mathbf{d}^\top \mathbf{Ad},$$

which exceeds $f(\mathbf{x}_*)$ unless $\mathbf{d} = \mathbf{0}$.

■

With $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$, this is least squares subject to a linear hypothesis $\mathbf{H}\boldsymbol{\beta} = \mathbf{h}$, the starting point of Chapter 11.

1.11.3 Derivatives with respect to matrices and scalars

Likelihoods for normal models involve $\log \det \boldsymbol{\Sigma}$ and $\text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S})$. For a scalar function f of an $m \times n$ matrix $\mathbf{W}$ with functionally independent entries, $\partial f / \partial \mathbf{W}$ is the $m \times n$ matrix with entries $\partial f / \partial w_{ij}$.

Proposition 1.11.4 (a) $\partial \text{tr}(\mathbf{A}\mathbf{W})/\partial \mathbf{W} = \mathbf{A}^\top$.

(b) If $\det \mathbf{W} > 0$, then $\partial \log \det \mathbf{W} / \partial \mathbf{W} = (\mathbf{W}^{-1})^\top$.

(c) If $\mathbf{A}(t)$ is a nonsingular matrix whose entries are differentiable functions of a scalar t , with derivative $\dot{\mathbf{A}} = \partial \mathbf{A} / \partial t$, then

$$\frac{\partial \mathbf{A}^{-1}}{\partial t} = -\mathbf{A}^{-1} \dot{\mathbf{A}} \mathbf{A}^{-1}, \quad \frac{\partial \log |\det \mathbf{A}|}{\partial t} = \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}).$$

Proof.

(a) $\text{tr}(\mathbf{A}\mathbf{W}) = \sum_{i,j} a_{ji} w_{ij}$. (c) Differentiate $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$ by the product rule to get $\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\partial \mathbf{A}^{-1}/\partial t = \mathbf{0}$. For the determinant, $\mathbf{A}(t+h) = \mathbf{A}(t)(\mathbf{I} + h\mathbf{A}^{-1}\dot{\mathbf{A}} + o(h))$. In the permutation expansion of $\det(\mathbf{I} + h\mathbf{C})$ only the diagonal product contributes terms of first order in h , so $\det(\mathbf{I} + h\mathbf{C}) = 1 + h \text{tr} \mathbf{C} + O(h^2)$. Hence $\det \mathbf{A}(t+h) = \det \mathbf{A}(t)(1 + h \text{tr}(\mathbf{A}^{-1}\dot{\mathbf{A}}) + o(h))$, and taking logarithms gives the formula. (b) Apply (c) with $\mathbf{A}(t) = \mathbf{W} + t\mathbf{e}_i\mathbf{e}_j^\top$. Then $\text{tr}(\mathbf{W}^{-1}\mathbf{e}_i\mathbf{e}_j^\top) = \mathbf{e}_j^\top \mathbf{W}^{-1} \mathbf{e}_i$, the (j, i) entry of $\mathbf{W}^{-1}$. ■

When $\mathbf{W}$ is constrained to be symmetric, w_{ij} and w_{ji} are the same variable, so the derivatives with respect to off-diagonal entries double. Some books (Renchner and Schaalje 2008) report formulas of the type $2\mathbf{W}^{-1} - \text{diag}(\mathbf{W}^{-1})$ for this reason. The simplest way to avoid the issue is to differentiate along a symmetric direction $\mathbf{W} + t\mathbf{E}$, as in (c). This is the convenient route for the normal likelihoods of Chapter 32. The listing checks the gradient formulas against central finite differences, and the script also checks part (c) and Proposition 1.11.2. The largest discrepancy in the listing is 3×10^{-9} , which is the size one expects from the finite-difference approximation.

```
import numpy as np
rng = np.random.default_rng(5)
p = 4
def num_grad(f, x, h=1e-6):
    """Central-difference gradient of a scalar function of a vector or matrix."""
    g = np.zeros_like(x)
    for idx in np.ndindex(x.shape):
        e = np.zeros_like(x)
        e[idx] = h
        g[idx] = (f(x + e) - f(x - e)) / (2 * h)
    return g

a = rng.normal(size=p)
A = rng.normal(size=(p, p)) # not symmetric
x = rng.normal(size=p)
checks = {
    "a'x": (num_grad(lambda v: a @ v, x), a),
    "x'Ax": (num_grad(lambda v: v @ A @ v, x), (A + A.T) @ x),
}
X = rng.normal(size=(20, p))
y = rng.normal(size=20)
b = rng.normal(size=p)
grad_ls = -2 * X.T @ (y - X @ b)
checks["||y - Xb||^2"] = (num_grad(lambda v: np.sum((y - X @ v) ** 2), b), grad_ls)
```

```

S = rng.normal(size=(p, p)); S = S @ S.T + p * np.eye(p)  # positive definite
checks["log det S"] = (num_grad(lambda W: np.linalg.slogdet(W)[1], S), np.linalg.inv
(S).T)
checks["tr(AS)"] = (num_grad(lambda W: np.trace(A @ W), S), A.T)
for name, (numeric, formula) in checks.items():
    print(f"{name:14s} max error {np.abs(numeric - formula).max():.1e}")

```

[Run this code online](#)

1.11.4 Exercises

1.11.4.1 A. Check your understanding

Exercise A1.

Observations fall in a clusters of size b , and $\Sigma = \sigma^2 \mathbf{I}_a \otimes \mathbf{E}_b$, where $\mathbf{E}_b = (1 - \rho)\mathbf{I}_b + \rho\mathbf{J}_b$. Find the eigenvalues, determinant and inverse of Σ using Proposition 1.10.1, Example (Equicorrelation) and Example (Eigenvalues of the equicorrelation matrix).

Exercise A2.

Use Proposition 1.11.2 to show that $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2$ with $\lambda > 0$ has the unique minimizer $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$, whatever the rank of $\mathbf{X}$.

1.11.4.2 B. Practice

Exercise B1.

Let $\mathbf{X}$ have full column rank, $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, and let $\mathbf{H}$ be $q \times p$ with full row rank. Show that the minimizer of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\mathbf{H}\mathbf{b} = \mathbf{h}$ is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}), \quad \mathbf{W} = \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top,$$

and that the increase in the residual sum of squares is $(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$.

Solution. Apply Proposition 1.11.3 with $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$. Since $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} + f(\mathbf{b})$ and $\mathbf{x}_0 = \hat{\boldsymbol{\beta}}$, this gives $\hat{\boldsymbol{\beta}}_H$. By completing the square as in Proposition 1.11.2, $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$ for every $\mathbf{b}$. With $\mathbf{b} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$, the increase is

$$(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}) = (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}).$$

This quadratic form is the numerator of the F statistic in Chapter 11.